MILITARY STANDARD

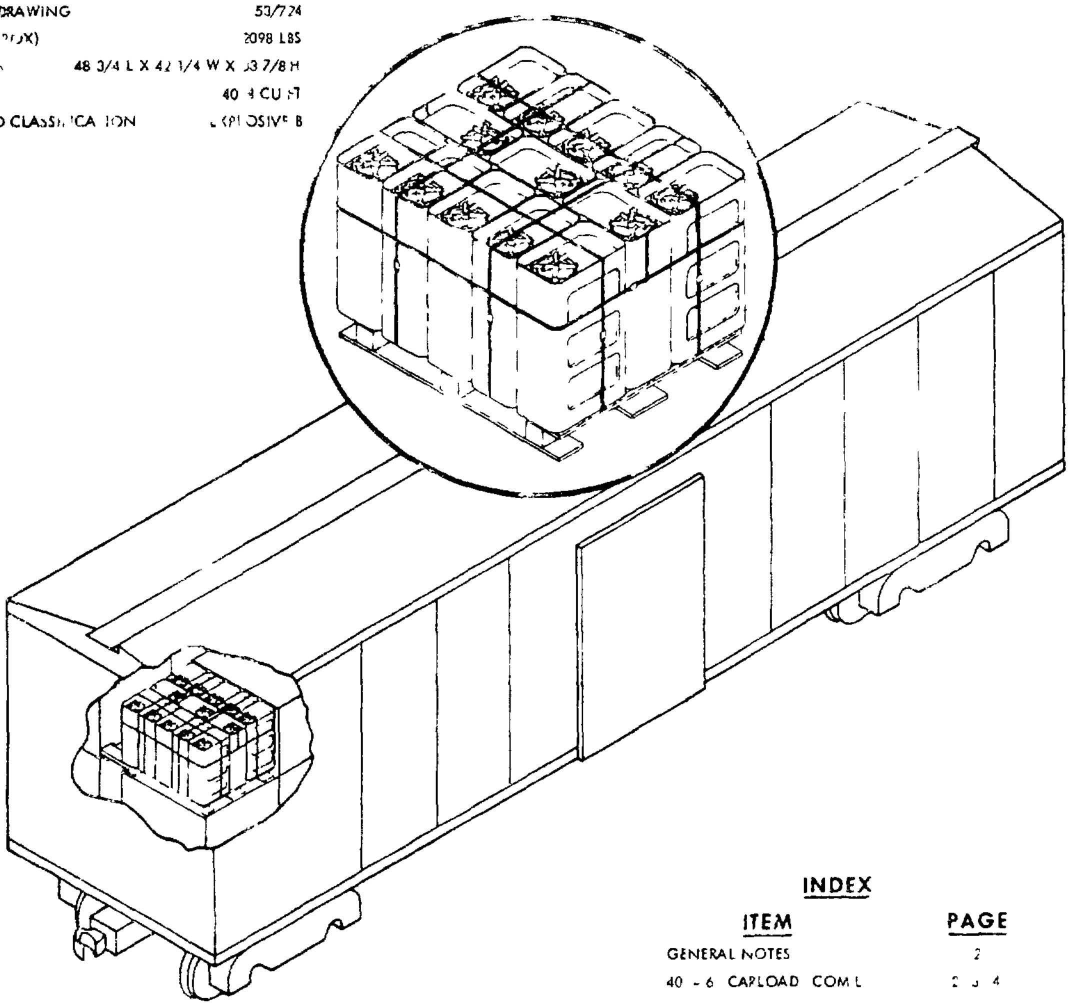
MIL-STD-1325-18
(NAVY)

RAILCAR LOADING OF HAZARDOUS MATERIALS
SMOKELESS POWDER FOR CANNON
IN PACKING BOX MK 7 AND MODS

1 OCTOBER 1976

UNIT LOAD DATA

UNIT LOAD DRAWING	53/7 24
WEIGHT (APPROX)	2098 LBS
DIMENSIONS	48 3/4 L X 42 1/4 W X 33 7/8 H
CUBE	40 4 CU FT
DOT HAZARD CLASSIFICATION	EXPLOSIVE B



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NOTES

1 UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES

2 FOR CROSS-REFERENCE TO ASSOCIATED PALLETIZING TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY)

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE

M S Kraus NWHC 9/22/76
TECHNICAL [illegible] (SEA) DATE

[illegible] 10/8/76
SEASYSCOM BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

7/9/76

ORIGINATOR SRP for 9/22/76 DATE
NAVAL WEAPONS HANDLING LABORATORY
[illegible] STA EARLE, NEW JERSEY

GENERAL NOTES

1 FOR GENERAL INFORMATION CONCERNING ORDERING INSPECTING AND PREPARING CARS AND FOR DUNNAGING MATERIALS, DESIGN AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) RAILCAR LOADING OF HAZARDOUS MATERIALS

2 WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT DF BOXCARS WITH MECHANICAL BRACING SHALL NOT BE USED FOR ITEMS HAVING EXPLOSIVE A OR B HAZARD CLASSIFICATION ALL STEEL BOXCARS SHALL NOT BE USED FOR SHIPMENT OF BULK EXPLOSIVES SUCH AS SMOKELESS POWDER BOXCARS WITH STEEL FLOORS AND WOOD SIDE AND END WALLS MUST HAVE THE STEEL FLOOR COVERED WITH EITHER DIMENSIONAL LUMBER (MINIMUM 1 INCH NOMINAL) 1/4 INCH PLYWOOD, 1/8 INCH HARDBOARD, OR SOLID FIBERBOARD CONFORMING TO FEDERAL SPECIFICATION PPP-F-320 TYPE SF (SOLID FIBERBOARD), CLASS - DOMESTIC, GRADE 175 OR CLASS-WEATHER-RESISTANT GRADE W6S (060 INCH THICKNESS) OR STRONGER THE COVERING MUST BE INSTALLED IN SUCH A MANNER THAT IT WILL NOT SHIFT DURING TRANSIT

3 LOADING PLANS SHOWN ARE FOR COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH

4 THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS

5 IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR

6 THE LOAD CONSISTS OF SMOKELESS POWDER FOR CANNON IN PACKING BOX MK 7 AND MODS PALLETIZED IN ACCORDANCE WITH WR-53/724

7 THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK

8 UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY)

9 ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971

10 APPLICABLE MATERIAL SPECIFICATIONS

DUNNAGE LUMBER - FED SPEC MM-L-751
NAILS - FED SPEC FF-N-105
STRAPPING - FED SPEC QQ-S-781 TYPE I HEAVY DUTY CLASS A DRY (UNLUBRICATED)
SEALS - FED SPEC QQ-S-781, STYLE III HEAVY DUTY

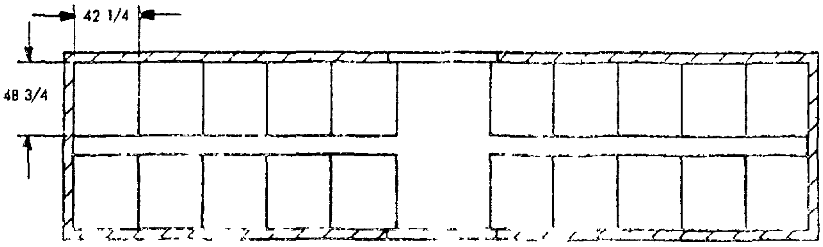
11 AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA CLOSE AND SEAL BOXCAR DOORS AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL I)

40 FT 6 IN BOXCAR, COMMERCIAL

1 THE CARLOAD CONSISTS OF 60 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING BOXCARS NOT HAVING THE LOAD CARRYING CAPACITY (LD LMT) FOR 60 UNIT LOADS MAY BE LOADED WITH A LESSER AMOUNT THE PROCEDURES SPECIFIED IN THIS DOCUMENT MUST BE FOLLOWED AS CLOSELY AS POSSIBLE IF A PARTIAL LAYER RESULTS THE PARTIAL LAYER OF LADING SHALL BE BRACED IN ACCORDANCE WITH MIL STD-1325-100 (NAVY) OR THE "VERTICAL MEMBER AND STRAPPING" PROCEDURE OR THE "KNEE BRACING" PROCEDURE SHOWN IN MIL-STD-1325-102 (NAVY) SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED

2 WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325-100 (NAVY) OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH THE "VERTICAL MEMBER AND STRAPPING" PROCEDURE OR THE "KNEE BRACING" PROCEDURE SHOWN IN MIL STD-1325-102 (NAVY) SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH SEE MIL STD 1325 (NAVY)

3 THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE



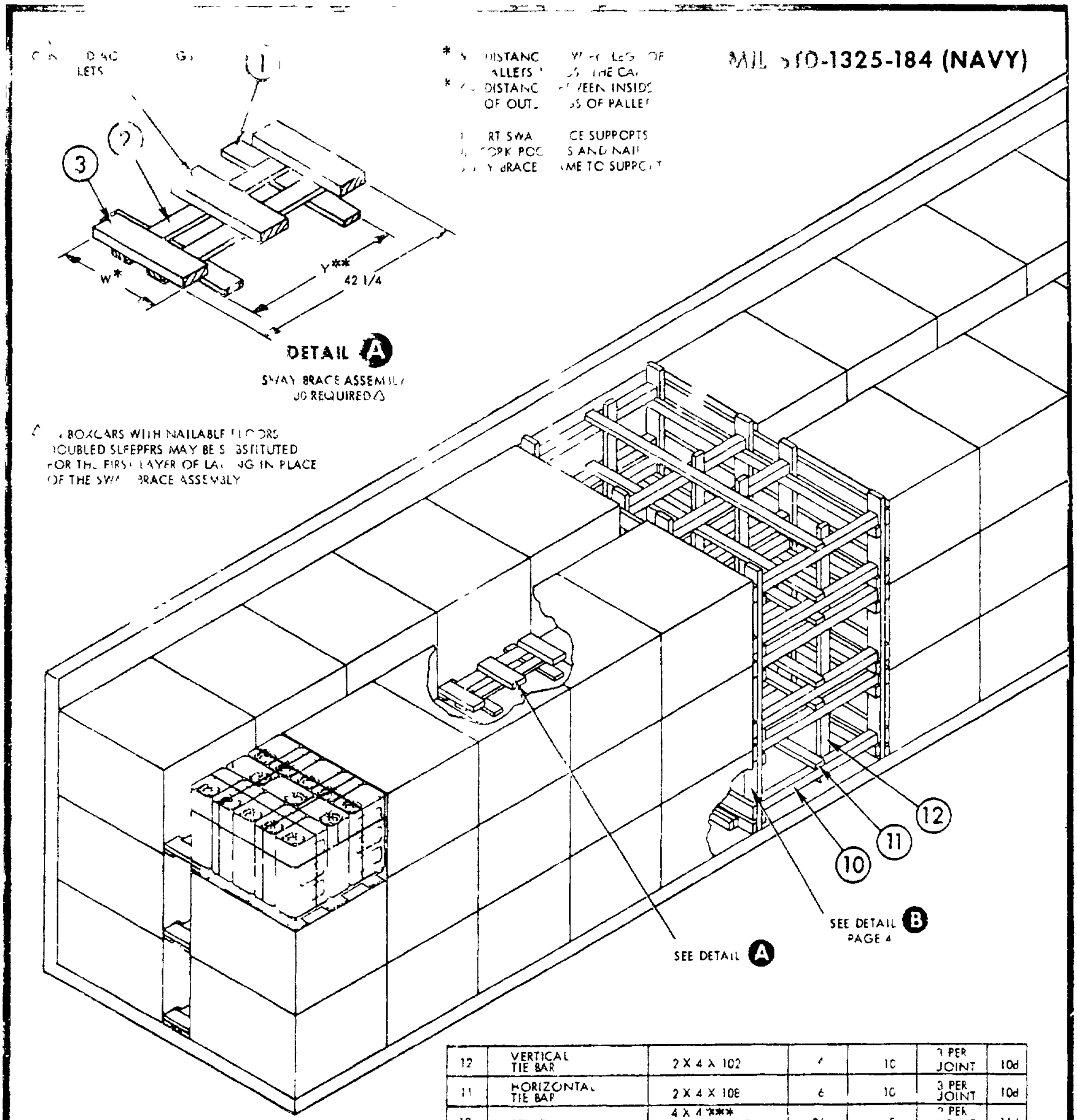
LOADING PLAN

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:03Z

* W DISTANC W LEG OF ALLETS S THE CA

** Y - DISTANC EEN INSIDE OF OUT S OF PALLET

1 RT SWA CE SUPPORTS ORK POC S AND NAI Y BRACE ME TO SUPPORT


C N D NC G LETS
1
2
3
W*
Y**
42 1/4
DETAIL A
SWAY BRACE ASSEMBLY
REQUIRED
BOXCARS WITH NAILABLE FLOORS DOUBLED SLEEPERS MAY BE SUBSTITUTED FOR THE FIRST LAYER OF LA NG IN PLACE OF THE SWAY BRACE ASSEMBLY
12
11
10
SEE DETAIL B
PAGE 4
SEE DETAIL A

12	VERTICAL TIE BAR	2 X 4 X 102	4	10	3 PER JOINT	10d
11	HORIZONTAL TIE BAR	2 X 4 X 108	6	10	3 PER JOINT	10d
10	STRUT	4 X 4 *** X WEDGE FIT	24	5	2 PER JOINT	16d
9	HOLDDOWN	2 X 4 X TO SUIT	4	8	4	10d
8	HOLDDOWN	2 X 4 X TO SUIT	4	7	4	10d
7	HOLDDOWN SPACER	2 X 4 X CAR WIDTH - 1	2	5	3 PER JOINT	10d
6	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH - 1	12	5	3 PER JOINT	10d
5	CENTER GATE VERTICAL	2 X 6 X 104	8	SEE 4	-	-
4	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH - 1	12	5	3 PER JOINT	10d
3	CROSS BRACE	2 X 6 X W*	90	SEE 2	-	-
2	SWAY BRACE LONGITUDINAL	1 X 4 X 42 1/4	60	1, 3	SEE NOTE	**
1	SWAY BRACE SUPPORT	1 X 4 X (W 8)*	60	SEE 2	-	-
	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	

LIST OF MATERIALS & NAILING DATA

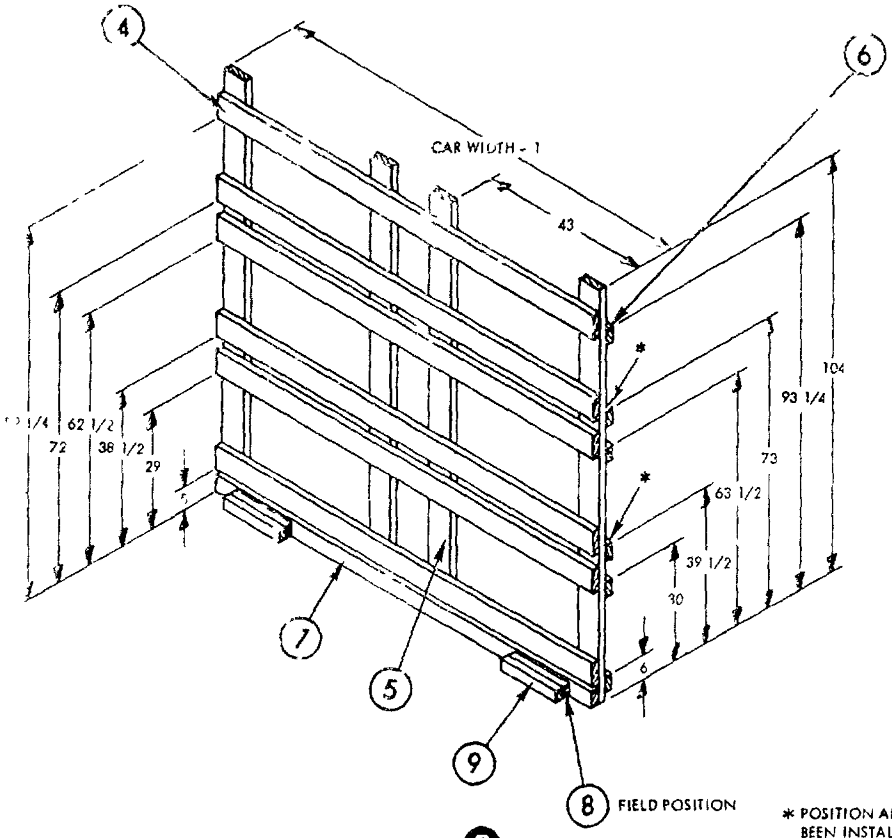
CARLOAD DATA

NUMBER OF UNIT LOADS	60
LOAD WEIGHT (APPROX)	125880 LBS
DUNNAGE WEIGHT (APPROX)	1651 LBS
CARLOAD WEIGHT (APPROX	127531 LBS

* NCE LEGS OF PALL IS STI

** PIECE 2 [illegible] TH [illegible] NAIL CLINCH D 3 PER [illegible], AND TO P [illegible] 14 N ILS, ER JOINT

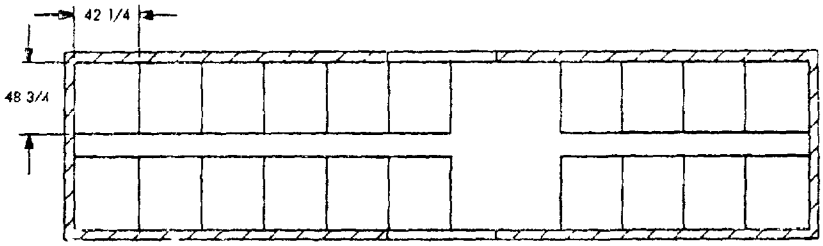
*** 6 STRU [illegible] D AND MINA D [illegible] F


4
6
CAR WIDTH - 1
43
104
93 1/4
73
63 1/2
39 1/2
30
6
*
*
1/4
62 1/2
72
38 1/2
29
5
7
5
9
8
FIELD POSITION

DETAIL Ⓑ

CENTER GATE
2 REQUIRED

* POSITION AFTER STRUTS HAVE
BEEN INSTALLED ON LOWER
STRUT CLEAT
(REQUIRED ONE GATE ONLY)



ALTERNATE LOADING PLAN

1 THIS ALTERNATE LOADING PLAN MAY BE USED WHEN THE LOAD LIMIT OF THE BOXCAR IS GREATER THAN 138,000 LBS

2 WHEN USING THIS ALTERNATE LOADING PLAN, DOORWAY PROTECTION IN ACCORDANCE WITH MIL STD-1325 (NAVY) MUST BE PROVIDED

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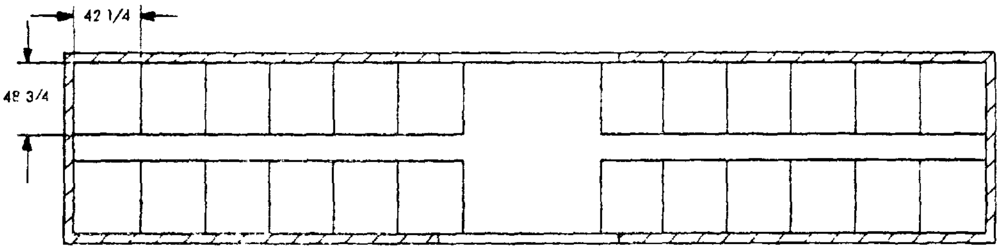
50 FT 6 IN BOXCAR, COMMERCIAL

1 THE CARLOAD CONSISTS OF 72 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING BOXCARS NOT HAVING THE LOAD CARRYING CAPACITY (LD LMT) FOR 72 UNIT LOADS MAY BE LOADED WITH A LESSER AMOUNT THE PROCEDURES SPECIFIED IN THIS DOCUMENT MUST BE FOLLOWED AS CLOSELY AS POSSIBLE IF A PARTIAL LAYER RESULTS THE PARTIAL LAYER OF LADING SHALL BE BRACED IN ACCORDANCE WITH MIL-STD-1325-100 (NAVY) OR THE "VERTICAL MEMBER AND STRAPPING" PROCEDURE OR THE "KNEE BRACING" PROCEDURE SHOWN IN MIL-STD-1325-102 (NAVY) SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED

2 WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325-100 (NAVY) OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH THE "VERTICAL MEMBER AND STRAPPING" PROCEDURE OR THE "KNEE BRACING" PROCEDURE SHOWN IN MIL-STD-1325-102(NAVY) SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH SEE MIL-STD-1325 (NAVY)

3 THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE

4 THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG DOORS PROVIDING STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA ARE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4 STEEL STRAPS PER STACK EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE CONSISTING OF A STRAPPING BOARD AND CROSS BRACE TWO PER STACK (SEE DETAIL C) INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS EACH STRAP IS SECURED TO STRAPPING BOARD WITH TWO 1 1/4' STAPLES DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR IN ADDITION TO UNITIZED STACKS IN THE DOORWAY AREA, BOXCARS HAVING A COMBINATION OF PLUG DOORS AND CONVENTIONAL SLIDING DOORS REQUIRE DIMENSIONAL LUMBER DOORWAY PROTECTION FOR THE SLIDING DOORS PLUG DOORS MUST BE SECURELY CLOSED AND WIRED TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER



LOADING PLAN

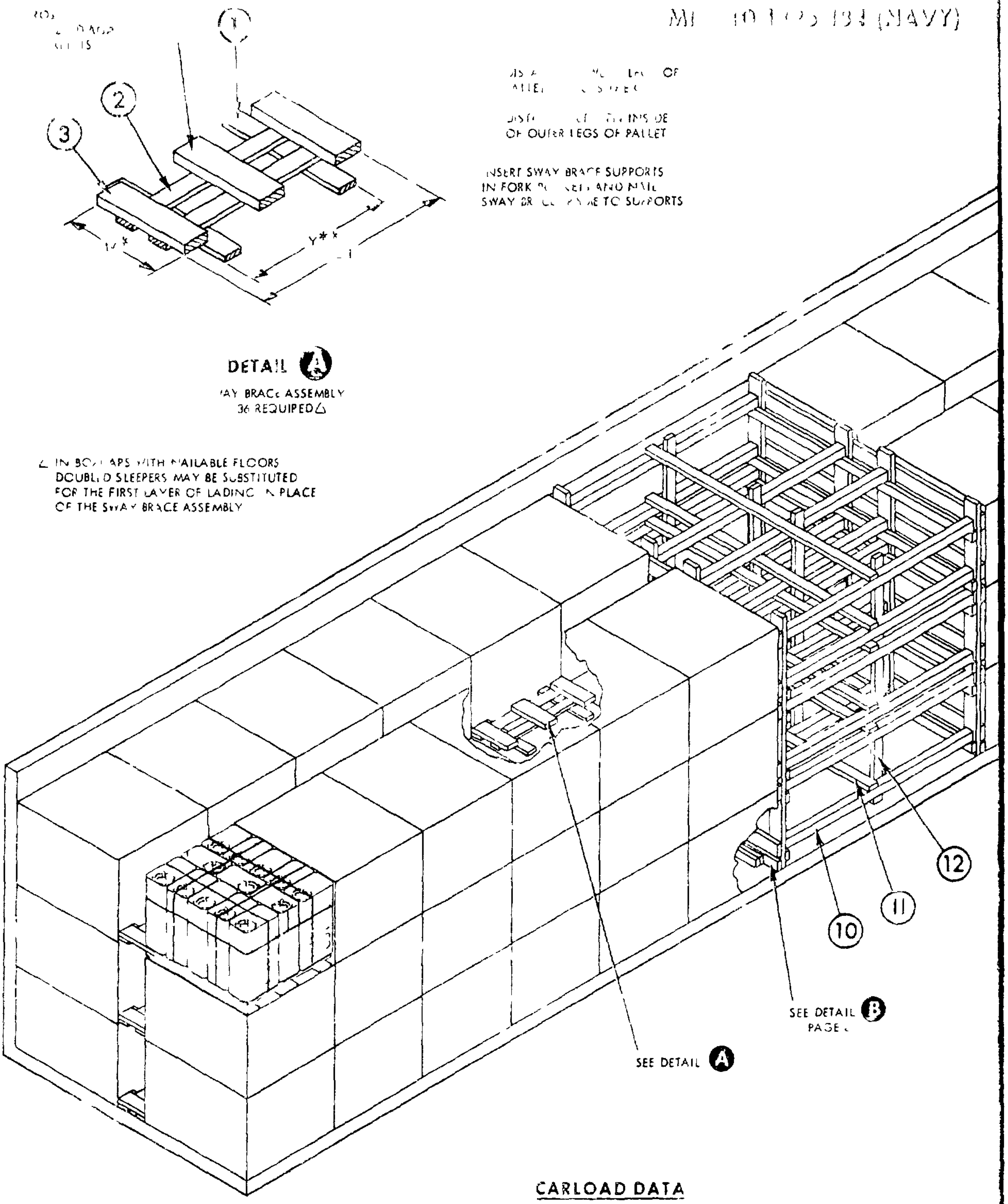
*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR

**- PIECE 2 NAILS TO PIECE 3 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 1 WITH 6d NAILS, 1 PER JOINT

***- 2 X 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 X 4's

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
12	VERTICAL TIE BAR	2 X 4 X 102	4	10	3 PER JOINT	10d
11	HORIZONTAL TIE BAR	2 X 4 X 108	6	10	3 PER JOINT	10d
10	STRUT	4 X 4*** X WEDGE FIT	24	5	2 PER JOINT	16d
9	HOLDDOWN	2 X 4 X TO SUIT	4	8	4	10d
8	HOLDDOWN	2 X 4 X TO SUIT	4	7	4	10d
7	HOLDDOWN SPACER	2 X 4 X CAR WIDTH - 1	2	5	3 PER JOINT	10d
6	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH - 1	12	5	3 PER JOINT	10d
5	CENTER GATE VERTICAL	2 X 6 X 104	8	SEE 4	-	-
4	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH - 1	12	5	3 PER JOINT	10d
3	CROSS BRACE	2 X 6 X W*	108	SEE 2	-	-
2	SWAY BRACE LONGITUDINAL	1 X 4 X 42 1/4	72	1, 3	SEE NOTE	**
1	SWAY BRACE SUPPORT	1 X 4 X (W + 8)*	72	SEE 2	-	-

LIST OF MATERIALS & NAILING DATA



DIS ... OF
PALLET ... SIZE

DIST ... INSIDE
OF OUTER LEGS OF PALLET

INSERT SWAY BRACE SUPPORTS
IN FORK ... AND NAIL
SWAY BR... TO SUPPORTS

DETAIL A
SWAY BRACE ASSEMBLY
36 REQUIRED△

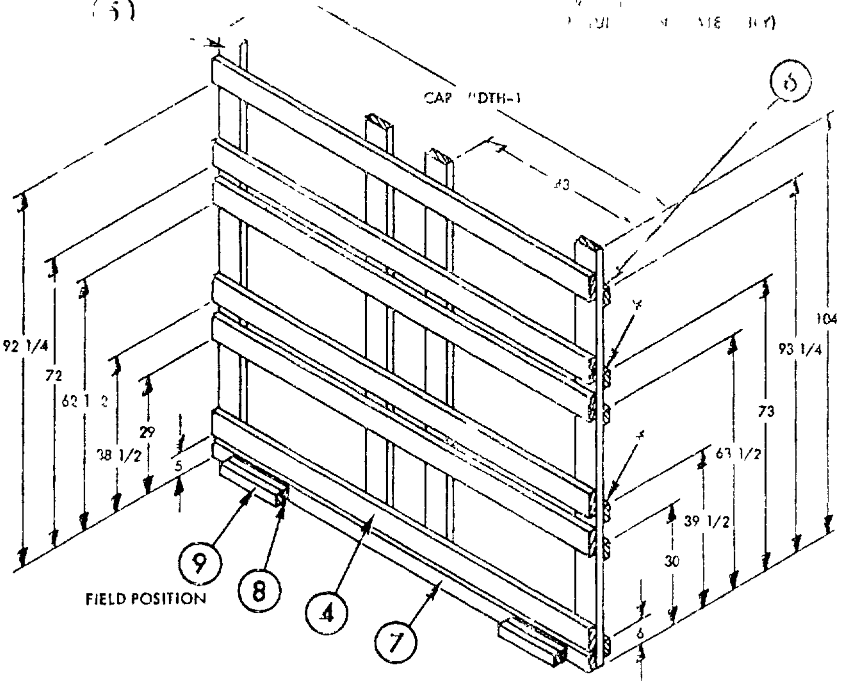
△ IN BOX... WITH NAILABLE FLOORS
DOUBLED SLEEPERS MAY BE SUBSTITUTED
FOR THE FIRST LAYER OF LADING IN PLACE
OF THE SWAY BRACE ASSEMBLY

SEE DETAIL B
PAGE ...

SEE DETAIL A

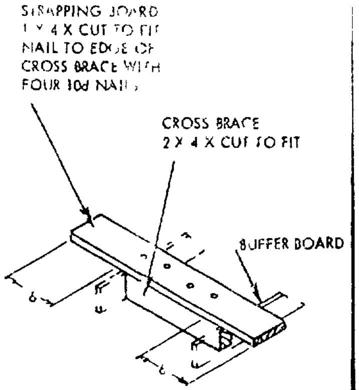
CARLOAD DATA

NUMBER OF UNIT LOADS	72
LOAD WEIGHT (APPROX	151 056 LBS
DUNNAGE WEIGHT (APPR	1,975 LBS
CARLOAD WEIG (APPR	53 031 [illegible]


[illegible] SHOULD HAVE
[illegible]
[illegible]
(5)
(6)
CAP WIDTH-1
43
92 1/4
72
62 1/2
38 1/2
29
5
104
93 1/4
73
63 1/2
39 1/2
30
6
9
8
4
7
FIELD POSITION

DETAIL B

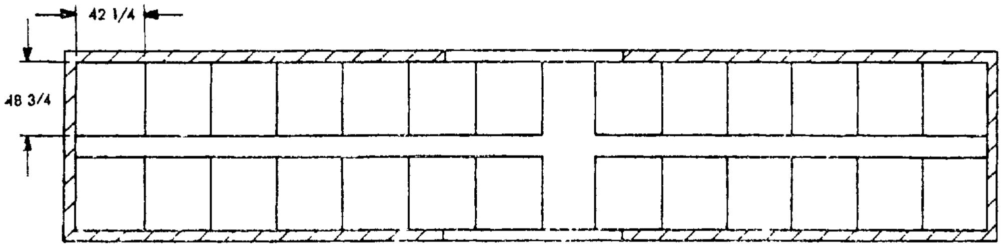
CENTER GATE
2 REQUIRED


STRAPPING BOARD
1 X 4 X CUT TO FIT
NAIL TO EDGE OF
CROSS BRACE WITH
FOUR 10d NAILS
CROSS BRACE
2 X 4 X CUT TO FIT
BUFFER BOARD
6
6

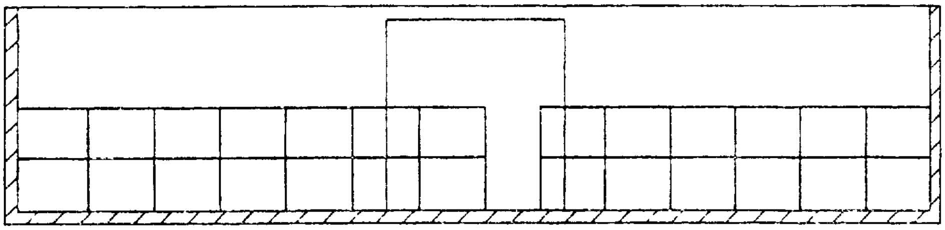
TO AVOID CONCENTRATED FORCES ON CONTAINERS CONNECT TWO TOP SWAY BRACES WITH BUFFER BOARDS (1 X 4 X 36) NAILED TO ENDS OF CROSS BRACES WITH TWO 10d NAILS EACH JOINT. TOP SWAY BRACES TO BE SPACED TO SUIT STRAPPING LOCATIONS

DETAIL C

TOP SWAY BRACE



ALTERNATE LOADING PLAN



ALTERNATE LOADING PLAN ELEVATION

1. THE ALTERNATE LOADING PLAN MAY BE USED WHEN THE LOAD LIMIT OF THE BOXCAR EXCEEDS 113,000 LBS BUT IS LESS THAN 153,500 LBS
2. WHEN USING THIS ALTERNATE LOADING PLAN, DOORWAY PROTECTION IN ACCORDANCE WITH MIL-STD-1325 (NAVY) MUST BE PROVIDED

CARLOAD DATA

NUMBER OF UNIT LOADS	52
LOAD WEIGHT (APPROX)	109096 LBS
DUNNAGE WEIGHT (APPROX)	1426 LBS
CARLOAD WEIGHT (APPROX)	110522 LBS

REVIEW ACTIVITY
NAVY - OS

PREPARING ACTIVITY
NAVY - OS
PROJECT (8140-N153)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-STD-1325-184 (Navy)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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